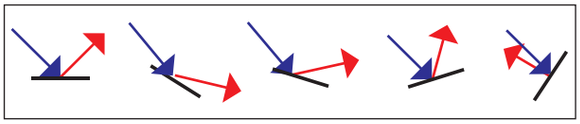
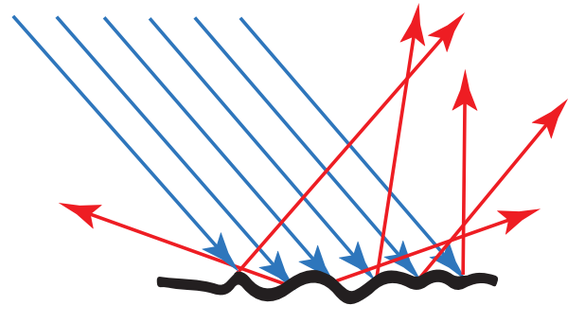
## Types of reflection

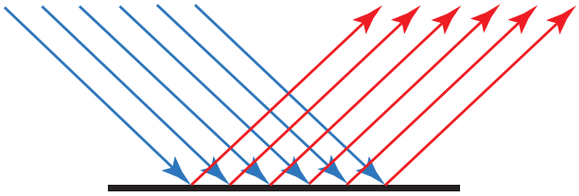
On a mirror we can see our image, but not on the wall. Why? Both the surface reflects light. Only because the reflected light comes to our eyes, we are able to see it. If the wall was not reflecting light, then we cannot see it.



We saw earlier that the light reflects off surfaces in a very predictable manner, in accordance with the law of reflection. The laws of reflection holds good for all surfaces irrespective of the shape. Vertical surfaces, angled surfaces, and even over the curved surfaces, the laws of reflection holds good. As long as we can draw the normal, perpendicular to the surface at the point can be drawn, the angle of incidence at that point will be equal to angle of reflection.



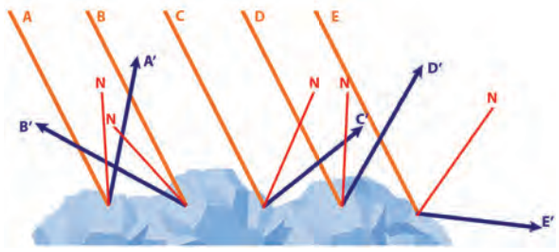
Diffuse Reflection
(Rough surface)



Specular Reflection
(Smooth surface)

The law of reflection is always observed regardless of the orientation of the surface. If the surface is smooth, and flat, all points on it have the normal in the same direction. Therefore a set of parallel rays striking the surface will be reflected at an angle, but the rays themselves will still remain parallel to each other.

However ,consider a surface which is not smooth. Such as the surface of a wall. What happen when the light ray hits the rough surface? Roughness of the wall means that each individual ray meets a surface which has a different orientation. The diagram below depicts the case. Five incident rays labelled as A, B, C, D and E approach a surface. The normal line at each point of incidence is shown in black and labelled with an N. In each case, the law of reflection is followed, resulting in five reflected rays labelled A', B', C', D', and E'. While the incident rays were parallel to each other, the reflected rays are going in different directions. The result is that the rays of light are incident upon the surface in a concentrated bundle and are diffused upon reflection.




Broadly, we can say that there are two types of reflection. If the surface is smooth then we have specular reflection. The parallel light rays striking the surface gets reflected, yet individual reflected rays remain parallel.

If the surface is rough, then we have diffused reflection. Light rays, after reflection go in many directions.

In fact during the day, our class room is illuminated by sunlight . Walls and floors are exposed to diffused reflection. Suppose walls



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